

```
%pip install -qU \
    langchain \
    langchain-core \
    langchain-groq \
    langchain-huggingface \
    langchain-pinecone \
    pinecone \
    sentence-transformers \
    pymupdf \
    groq \
    pandas \
    tabulate

%pip uninstall -y -q torchvision torchaudio
```

WARNING: Skipping torchvision as it is not installed.
 WARNING: Skipping torchaudio as it is not installed.

```
import os
import io
import time
import base64
from pathlib import Path
from getpass import getpass

# import fitz
import pymupdf as fitz
import pandas as pd

from PIL import Image
from IPython.display import display

from groq import Groq

from pinecone import Pinecone, ServerlessSpec

from langchain_core.documents import Document
from langchain_core.prompts import ChatPromptTemplate
from langchain_core.output_parsers import StrOutputParser
from langchain_groq import ChatGroq
from langchain_huggingface import HuggingFaceEmbeddings
from langchain_pinecone import PineconeVectorStore
```

```
PDF_PATH = Path("NovaCore_Multimodal_Company_Report_2026.pdf")

# Fallback for this generated environment.
if not PDF_PATH.exists():
    fallback = Path("/mnt/data/NovaCore_Multimodal_Company_Report_2026.pdf")
    if fallback.exists():
        PDF_PATH = fallback

assert PDF_PATH.exists(), (
    "PDF not found. Keep NovaCore_Multimodal_Company_Report_2026.pdf "
    "in the same folder as this notebook."
)
```

```
PDF_PATH
```

```
PosixPath('NovaCore_Multimodal_Company_Report_2026.pdf')
```

```
TEXT_MODEL = "openai/gpt-oss-20b"
VISION_MODEL = "qwen/qwen3.8-27b"
EMBEDDING_MODEL = "sentence-transformers/all-MiniLM-L6-v2"

PINECONE_INDEX_NAME = "novacore-multimodal-rag"
PINECONE_NAMESPACE = "fy2026-demo"

IMAGE_DIR = Path("novacore_extracted_images")
IMAGE_DIR.mkdir(exist_ok=True)

print("PDF:", PDF_PATH)
print("Pinecone index:", PINECONE_INDEX_NAME)
print("Pinecone namespace:", PINECONE_NAMESPACE)

PDF: NovaCore_Multimodal_Company_Report_2026.pdf
Pinecone index: novacore-multimodal-rag
Pinecone namespace: fy2026-demo
```

```

from google.colab import userdata
# Import ChatGroq and TEXT_MODEL based on your setup

# Securely fetch keys from Colab Secrets
try:
    os.environ["GROQ_API_KEY"] = userdata.get('GROQ_API_KEY')
    os.environ["PINECONE_API_KEY"] = userdata.get('PINECONE_API_KEY')
except userdata.SecretNotFoundError:
    raise ValueError("API Keys not found! Please add GROQ_API_KEY and PINECONE_API_KEY to your Colab Secrets (Key icon).")

# Initialize clients using the secured environment variables
groq_client = Groq(api_key=os.environ["GROQ_API_KEY"])

text_llm = ChatGroq(
    model=TEXT_MODEL,
    temperature=0,
)

print("API clients are ready.")

```

API clients are ready.

```

def image_to_data_uri(image_path):
    image_path = Path(image_path)

    with Image.open(image_path) as img:
        img = img.convert("RGB")
        img.thumbnail((1600, 1600))

        buffer = io.BytesIO()
        img.save(buffer, format="JPEG", quality=85)

    encoded = base64.b64encode(buffer.getvalue()).decode("utf-8")
    return f"data:image/jpeg;base64,{encoded}"

```

```

def summarize_visual(image_path, page_number):
    image_data = image_to_data_uri(image_path)

    prompt = f"""
This visual was extracted from page {page_number} of the NovaCore FY2026 company report.

Describe the useful business information visible in the visual.

If it is a chart or graph:
- mention important values
- mention highest/lowest values
- mention the main trend

If it is a diagram:
- identify important components
- explain the flow or relationships

If it is a normal business image:
- describe the useful factual information

Keep the description concise and factual.
"""

    response = groq_client.chat.completions.create(
        model=VISION_MODEL,
        messages=[
            {
                "role": "user",
                "content": [
                    {"type": "text", "text": prompt},
                    {
                        "type": "image_url",
                        "image_url": {"url": image_data},
                    },
                ],
            },
        ],
        temperature=0,
        max_completion_tokens=500,
    )

    return response.choices[0].message.content.strip()

```

```

def extract_multimodal_documents(pdf_path):

```

```

pdf = fitz.open(str(pdf_path))

documents = []
extracted_xrefs = set()

for page_index in range(len(pdf)):
    page = pdf[page_index]
    page_number = page_index + 1

    print(f"Processing page {page_number}...")

    # -----
    # 1. TEXT
    # -----
    text = page.get_text("text").strip()

    if text:
        documents.append(
            Document(
                page_content=text,
                metadata={
                    "page": page_number,
                    "modality": "text",
                    "source": pdf_path.name,
                },
            )
        )

    # -----
    # 2. TABLES
    # -----
    try:
        tables = page.find_tables().tables

        for table_number, table in enumerate(tables, start=1):
            df = table.to_pandas()

            if not df.empty:
                table_text = df.to_markdown(index=False)

                documents.append(
                    Document(
                        page_content=table_text,
                        metadata={
                            "page": page_number,
                            "modality": "table",
                            "table_number": table_number,
                            "source": pdf_path.name,
                        },
                    )
                )

    except Exception as error:
        print("Table extraction warning:", error)

    # -----
    # 3. IMAGES / CHARTS / DIAGRAMS
    # -----
    for image_number, image_info in enumerate(
        page.get_images(full=True),
        start=1,
    ):
        xref = image_info[0]

        # Do not process an identical embedded image twice.
        if xref in extracted_xrefs:
            continue

        extracted_xrefs.add(xref)

        image_data = pdf.extract_image(xref)
        image_bytes = image_data["image"]
        image_extension = image_data["ext"]

        image_path = (
            IMAGE_DIR
            / f"page_{page_number}_image_{image_number}.{image_extension}"
        )

        image_path.write_bytes(image_bytes)

    try:
        summary = summarize_visual(

```

```
summary = summary_image_visualize(
    image_path=image_path,
    page_number=page_number,
)

documents.append(
    Document(
        page_content=summary,
        metadata={
            "page": page_number,
            "modality": "visual",
            "image_path": str(image_path),
            "source": pdf_path.name,
        },
    )
)

except Exception as error:
    print(
        f"Vision warning on page {page_number}:",
        error,
    )

pdf.close()
return documents

documents = extract_multimodal_documents(PDF_PATH)

print()
print("Total LangChain documents:", len(documents))
```

Processing page 1...
Consider using the pymupdf_layout package for a greatly improved page layout analysis.
Processing page 2...
Processing page 3...
Processing page 4...
Processing page 5...
Processing page 6...
Processing page 7...
Processing page 8...
Processing page 9...

Total LangChain documents: 24

documents

```
[Document(metadata={'page': 1, 'modality': 'text', 'source': 'NovaCore_Multimodal_Company_Report_2026.pdf'},
page_content='NovaCore Systems Ltd. - Fictional company report created for a multimodal RAG demonstration\nPage
1\nNOVACORE SYSTEMS LTD.\nFY2026 BUSINESS PERFORMANCE & \nOPERATIONS REPORT\nA realistic fictional company document
designed for multimodal RAG testing\nHeadquarters\nSingapore\nEmployees\n1,480\n2026 Revenue\n$132.0M\nOperating
Margin\n18.4%\nPrimary Sector\nIndustrial AI & IoT\nMarkets Served\n42 countries\nDemo note: All company names, metrics,
people, and events in this report are fictional. The document intentionally mixes text, tables,\ncharts, and embedded
images so it can be used to demonstrate multimodal retrieval and question answering.'),
 Document(metadata={'page': 1, 'modality': 'table', 'table_number': 1, 'source':
'NovaCore_Multimodal_Company_Report_2026.pdf'}, page_content='| Headquarters | Singapore | Employees |
1,480 | \n|:-----|:-----|:-----|:-----|\n| 2026 Revenue | $132.0M
| Operating Margin | 18.4% | \n| Primary Sector | Industrial AI & IoT | Markets Served | 42 countries |'),
 Document(metadata={'page': 1, 'modality': 'visual', 'image_path': 'novacore_extracted_images/page_1_image_1.png',
'source': 'NovaCore_Multimodal_Company_Report_2026.pdf'}, page_content='Based on the visual from the NovaCore FY2026
company report, here is the useful business information:\n\n**Facility Overview:**\n* **Name:** Penang Smart Assembly
Facility\n* **Opening Date:** 2024\n* **Size:** 18,000 m²\n* **Certification:** ISO 9001 certified\n\n**Key
Performance Indicators (2026):**\n* **Facility Utilization:** 82%\n* **Solar Contribution to Electricity:**
34%\n\n**Visual Components:**\n* The image depicts a modern industrial building with solar panels installed on the
roof, visually reinforcing the "Smart" and sustainability aspects mentioned in the text.\n* A delivery truck is shown
at the loading dock, indicating active logistics and operations.'),
 Document(metadata={'page': 2, 'modality': 'text', 'source': 'NovaCore_Multimodal_Company_Report_2026.pdf'},
page_content='NovaCore Systems Ltd. - Fictional company report created for a multimodal RAG demonstration\nPage 2\n1.
Executive Overview\nNarrative text section - useful for text retrieval and grounded summarization\nNovaCore Systems Ltd.
develops industrial AI gateways, computer-vision appliances, and connected sensor hubs\nfor manufacturing, logistics, and
energy companies. During FY2026, the company focused on three priorities:\nexanding enterprise deployments in Europe,
improving the gross margin of its flagship AtlasEdge product line,\nand shortening post-deployment support response
times.\nFY2026 revenue reached USD 132.0 million, up from USD 97.5 million in FY2025. Europe became the\nfastest-growing
region, supported by automotive and warehouse automation projects. The AtlasEdge X1 remained\nthe highest-margin product,
while VisionHub Pro generated the largest number of new enterprise pilots.\nManagement approved a further USD 11.5
million investment in the Penang assembly facility for test automation\nand energy-efficiency upgrades.\n\nThe company
ended the year with 1,480 employees. Engineering represented 38% of headcount, operations 27%,\nsales and customer
success 22%, and corporate functions 13%. Customer satisfaction improved as average\nsupport resolution time declined
steadily through August.\n\n2026 Strategic Priorities\n\nPriority\n\n2026 target\n\nYear-end status\n\nEuropean expansion\n>25% YoY
regional revenue growth\n\nAchieved: 31%\n\nSupport speed\n\nAverage resolution <9 hours\n\nAchieved in July; 8.1 hours in
August\n\nFacility utilization\n\n80%+ Penang utilization\n\nAchieved: 82%\n\nProduct profitability\n\nFlagship gross margin
>44%\n\nAchieved: AtlasEdge X1 at 46%\n\nCross-modal clue: the exact electricity mix of the Penang facility appears later in
a chart, while the\nsupply-chain control point appears inside an image diagram.'),
 Document(metadata={'page': 2, 'modality': 'table', 'table_number': 1, 'source':
'NovaCore_Multimodal_Company_Report_2026.pdf'}, page_content='| Priority | 2026 target
| Year-end status | \n|:-----|:-----|:-----|:-----|')]
```

```

-----|\\n| European expansion      | >25% YoY regional revenue growth | Achieved: 31%
|\\n| Support speed                  | Average resolution <9 hours      | Achieved in July; 8.1 hours in August |\\n| Facility
utilization | 80%+ Penang utilization | Achieved: 82%                    |\\n| Product profitability |
Flagship gross margin >44%          | Achieved: AtlasEdge X1 at 46%    |'),
Document(metadata={'page': 3, 'modality': 'text', 'source': 'NovaCore_Multimodal_Company_Report_2026.pdf'},
page_content='NovaCore Systems Ltd. - Fictional company report created for a multimodal RAG demonstration\\nPage 3\\n2.
Financial Performance\\nTable + line graph\\nMetric\\nFY2025\\nFY2026\\nChange\\nRevenue\\n$97.5M\\n$132.0M\\n+35.4%\\nGross
profit\\n$39.7M\\n$56.4M\\n+42.1%\\nOperating income\\n$13.8M\\n$24.3M\\n+76.1%\\nOperating margin\\n14.2%\\n18.4%\\n+4.2 pts\\nR&D
investment\\n$12.1M\\n$17.9M\\n+47.9%\\nFigure 1. Quarterly revenue increased every quarter across the displayed two-year
period. The chart contains the detailed\\nquarter-by-quarter pattern, including the FY2026 peak.\\nManagement attributed
the margin expansion mainly to higher AtlasEdge X1 volume, improved component\\npurchasing terms, and better manufacturing
yield at the Penang facility. Foreign-exchange movements had a\\nmodest negative effect on reported European revenue.\\n'),
Document(metadata={'page': 3, 'modality': 'table', 'table_number': 1, 'source':
'NovaCore_Multimodal_Company_Report_2026.pdf'}, page_content='| Metric          | FY2025   | FY2026   | Change |\\n|:--
-----|:-----|:-----|:-----|\\n| Revenue            | $97.5M   | $132.0M  | +35.4% |\\n| Gross profit
| $39.7M   | $56.4M   | +42.1%   |\\n| Operating income | $13.8M    | $24.3M    | +76.1%   |\\n| Operating margin | 14.2%
| 18.4%    | +4.2 pts |\\n| R&D investment   | $12.1M   | $17.9M   | +47.9% |'),
Document(metadata={'page': 3, 'modality': 'visual', 'image_path': 'novacore_extracted_images/page_3_image_1.png',
'source': 'NovaCore_Multimodal_Company_Report_2026.pdf'}). page_content='Based on the visual provided. here is the

```

```

rows = []

for doc in documents:
    rows.append(
        {
            "page": doc.metadata["page"],
            "modality": doc.metadata["modality"],
            "preview": doc.page_content[:140].replace("\\n", " "),
        }
    )

summary_df = pd.DataFrame(rows)

display(summary_df)

print("Documents by modality:")
display(summary_df["modality"].value_counts())

```

	page	modality	preview
0	1	text	NovaCore Systems Ltd. - Fictional company repo...
1	1	table	Headquarters Singapore Emplo...
2	1	visual	Based on the visual from the NovaCore FY2026 c...
3	2	text	NovaCore Systems Ltd. - Fictional company repo...
4	2	table	Priority 2026 target ...
5	3	text	NovaCore Systems Ltd. - Fictional company repo...
6	3	table	Metric FY2025 FY2026 Cha...
7	3	visual	Based on the visual provided, here is the desc...
8	4	text	NovaCore Systems Ltd. - Fictional company repo...
9	4	table	Region 2026 revenue YoY growth ...
10	4	visual	Based on the visual provided, here is the desc...
11	5	text	NovaCore Systems Ltd. - Fictional company repo...
12	5	table	Product 2026 units Average sell...
13	5	visual	Based on the visual provided, here is the usef...
14	6	text	NovaCore Systems Ltd. - Fictional company repo...
15	6	visual	Based on the visual provided, here is the desc...
16	7	text	NovaCore Systems Ltd. - Fictional company repo...
17	7	table	Severity Response target Resoluti...
18	7	visual	Based on the visual provided, here is the desc...
19	8	text	NovaCore Systems Ltd. - Fictional company repo...
20	8	table	Metric 2025 ...
21	8	visual	Based on the visual provided, here is the desc...
22	9	text	NovaCore Systems Ltd. - Fictional company repo...
23	9	table	Risk Likeliho...

Documents by modality:
modality
text 9
table 8
visual 7
Name: count, dtype: int64

```
from langchain_huggingface import HuggingFaceEmbeddings

# Define your model string (e.g., "all-MiniLM-L6-v2")
EMBEDDING_MODEL = "sentence-transformers/all-MiniLM-L6-v2"

embeddings = HuggingFaceEmbeddings(
    model_name=EMBEDDING_MODEL,
    encode_kwargs={"normalize_embeddings": True},
)

# Detect the exact vector dimension automatically.
EMBEDDING_DIMENSION = len(embeddings.embed_query("dimension check"))

print("Embedding model:", EMBEDDING_MODEL)
print("Embedding dimension:", EMBEDDING_DIMENSION)
```

Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and fast
 WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN t
 modules.json: 100% 349/349 [00:00<00:00, 27.4kB/s]
 config_sentence_transformers.json: 100% 116/116 [00:00<00:00, 9.72kB/s]
 README.md: 100% 10.5k/10.5k [00:00<00:00, 914kB/s]
 sentence_bert_config.json: 100% 53.0/53.0 [00:00<00:00, 3.62kB/s]
 config.json: 100% 612/612 [00:00<00:00, 58.7kB/s]
 model.safetensors: reconstructing file: 100% 90.9MB / 90.9MB, 8.79MB/s
 model.safetensors: downloading bytes: 85.0MB, 8.13MB/s
 Loading weights: 100% 103/103 [00:00<00:00, 2072.02it/s]
 tokenizer_config.json: 100% 350/350 [00:00<00:00, 30.9kB/s]
 vocab.txt: 100% 232k/232k [00:00<00:00, 10.6MB/s]
 tokenizer.json: 100% 466k/466k [00:00<00:00, 22.3MB/s]
 special_tokens_map.json: 100% 112/112 [00:00<00:00, 6.97kB/s]
 config.json: 100% 190/190 [00:00<00:00, 15.7kB/s]
 Embedding model: sentence-transformers/all-MiniLM-L6-v2
 Embedding dimension: 384

```
!pip install sentence-transformers
embeddings = HuggingFaceEmbeddings(
    model_name=EMBEDDING_MODEL,
    encode_kwargs={"normalize_embeddings": True},
)

# Detect the exact vector dimension automatically.
EMBEDDING_DIMENSION = len(
    embeddings.embed_query("dimension check")
)

print("Embedding model:", EMBEDDING_MODEL)
print("Embedding dimension:", EMBEDDING_DIMENSION)
```

```

Requirement already satisfied: sentence-transformers in /usr/local/lib/python3.13/dist-packages (6.1.0)
Requirement already satisfied: transformers<6.0.0,>=5.0.0 in /usr/local/lib/python3.13/dist-packages (from sentence-transformers) (4.41.0)
Requirement already satisfied: tokenizers>=0.19 in /usr/local/lib/python3.13/dist-packages (from sentence-transformers) (0.20.1)
Requirement already satisfied: huggingface-hub<2.0.0,>=1.3.0 in /usr/local/lib/python3.13/dist-packages (from sentence-transformers) (1.3.0)
Requirement already satisfied: torch>=2.2 in /usr/local/lib/python3.13/dist-packages (from sentence-transformers) (2.11.0+cpu)
Requirement already satisfied: numpy>=1.24.0 in /usr/local/lib/python3.13/dist-packages (from sentence-transformers) (2.1.3)
Requirement already satisfied: scikit-learn>=1.1.0 in /usr/local/lib/python3.13/dist-packages (from sentence-transformers) (1.5.2)
Requirement already satisfied: scipy>=1.0.0 in /usr/local/lib/python3.13/dist-packages (from sentence-transformers) (1.16.3)
Requirement already satisfied: typing_extensions>=4.10.0 in /usr/local/lib/python3.13/dist-packages (from sentence-transformers) (4.12.0)
Requirement already satisfied: tqdm>=4.0.0 in /usr/local/lib/python3.13/dist-packages (from sentence-transformers) (4.67.3)
Requirement already satisfied: click<9.0.0,>=8.4.2 in /usr/local/lib/python3.13/dist-packages (from huggingface-hub<2.0.0,>=1.3.0) (8.1.8)
Requirement already satisfied: filelock>=3.10.0 in /usr/local/lib/python3.13/dist-packages (from huggingface-hub<2.0.0,>=1.3.0) (3.16.1)
Requirement already satisfied: fsspec>=2023.5.0 in /usr/local/lib/python3.13/dist-packages (from huggingface-hub<2.0.0,>=1.3.0) (2025.3.0)
Requirement already satisfied: hf-xet<2.0.0,>=1.5.2 in /usr/local/lib/python3.13/dist-packages (from huggingface-hub<2.0.0,>=1.3.0) (1.2.0)
Requirement already satisfied: httpx<1,>=0.23.0 in /usr/local/lib/python3.13/dist-packages (from huggingface-hub<2.0.0,>=1.3.0) (0.27.0)
Requirement already satisfied: packaging>=20.9 in /usr/local/lib/python3.13/dist-packages (from huggingface-hub<2.0.0,>=1.3.0) (25.0)
Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.13/dist-packages (from huggingface-hub<2.0.0,>=1.3.0) (6.0.2)
Requirement already satisfied: joblib>=1.2.0 in /usr/local/lib/python3.13/dist-packages (from scikit-learn>=1.1.0->sentence-transformers) (1.4.2)
Requirement already satisfied: threadpoolctl>=3.1.0 in /usr/local/lib/python3.13/dist-packages (from scikit-learn>=1.1.0->sentence-transformers) (3.5.0)
Requirement already satisfied: setuptools>=82 in /usr/local/lib/python3.13/dist-packages (from torch>=2.2->sentence-transformers) (75.8.2)
Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.13/dist-packages (from torch>=2.2->sentence-transformers) (1.13.3)
Requirement already satisfied: networkx>=2.5.1 in /usr/local/lib/python3.13/dist-packages (from torch>=2.2->sentence-transformers) (3.4.2)
Requirement already satisfied: Jinja2 in /usr/local/lib/python3.13/dist-packages (from torch>=2.2->sentence-transformers) (3.1.4)
Requirement already satisfied: regex>=2025.10.22 in /usr/local/lib/python3.13/dist-packages (from transformers<6.0.0,>=5.0.0) (2025.10.22)
Requirement already satisfied: typer in /usr/local/lib/python3.13/dist-packages (from transformers<6.0.0,>=5.0.0->sentence-transformers) (0.12.5)
Requirement already satisfied: safetensors>=0.8.0 in /usr/local/lib/python3.13/dist-packages (from transformers<6.0.0,>=5.0.0) (0.5.2)
Requirement already satisfied: anyio in /usr/local/lib/python3.13/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0.0) (4.8.0)
Requirement already satisfied: certifi in /usr/local/lib/python3.13/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0.0) (2025.8.3)
Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.13/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0.0) (1.0.7)
Requirement already satisfied: idna in /usr/local/lib/python3.13/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0.0) (3.10)
Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.13/dist-packages (from httpcore==1.*->httpx<1,>=0.23.0->huggingface-hub<2.0.0) (0.14.0)
Requirement already satisfied: cloudpickle>=3.0 in /usr/local/lib/python3.13/dist-packages (from joblib>=1.2.0->scikit-learn>=1.1.0->sentence-transformers) (3.1.1)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.13/dist-packages (from sympy>=1.13.3->torch>=2.2->sentence-transformers) (1.3.0)
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.13/dist-packages (from Jinja2->torch>=2.2->sentence-transformers) (3.0.2)
Requirement already satisfied: shellingham>=1.3.0 in /usr/local/lib/python3.13/dist-packages (from typer->transformers<6.0.0,>=5.0.0->sentence-transformers) (1.5.4)
Requirement already satisfied: rich>=13.8.0 in /usr/local/lib/python3.13/dist-packages (from typer->transformers<6.0.0,>=5.0.0->sentence-transformers) (13.9.4)
Requirement already satisfied: annotated-doc>=0.0.2 in /usr/local/lib/python3.13/dist-packages (from typer->transformers<6.0.0,>=5.0.0->sentence-transformers) (0.0.2)
Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.13/dist-packages (from rich>=13.8.0->typer->sentence-transformers) (3.0.0)
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.13/dist-packages (from rich>=13.8.0->typer->sentence-transformers) (2.18.1)
Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.13/dist-packages (from markdown-it-py>=2.2.0->rich>=13.8.0->typer->sentence-transformers) (0.1.2)

```

Loading weights: 100%

103/103 [00:00<00:00, 2876.62it/s]

Embedding model: sentence-transformers/all-MiniLM-L6-v2
 Embedding dimension: 384

```

pc = Pinecone(
    api_key=os.environ["PINECONE_API_KEY"]
)

if not pc.has_index(PINECONE_INDEX_NAME):
    print("Creating Pinecone index...")

    pc.create_index(
        name=PINECONE_INDEX_NAME,
        dimension=EMBEDDING_DIMENSION,
        metric="cosine",
        spec=ServerlessSpec(
            cloud="aws",
            region="us-east-1",
        ),
    )

    # Wait until Pinecone reports that the index is ready.
    while not pc.describe_index(PINECONE_INDEX_NAME).status["ready"]:
        print("Waiting for Pinecone index...")
        time.sleep(2)

    print("Pinecone index created.")
else:
    print("Pinecone index already exists.")

index = pc.Index(PINECONE_INDEX_NAME)

print("Connected to:", PINECONE_INDEX_NAME)

```

Pinecone index already exists.
 Connected to: novacore-multimodal-rag

```

try:
    index.delete(
        delete_all=True,
        namespace=PINECONE_NAMESPACE,
    )

    print("Cleared namespace:", PINECONE_NAMESPACE)

```

```
except Exception as error:
    print(
        "Namespace was probably empty. Continuing...",
        error,
    )
```

Cleared namespace: fy2026-demo

```
vectorstore = PineconeVectorStore(
    index_name=PINECONE_INDEX_NAME,
    embedding=embeddings,
    namespace=PINECONE_NAMESPACE,
)

vectorstore.add_documents(documents)

print("Documents uploaded to Pinecone.")
```

Documents uploaded to Pinecone.

```
retriever = vectorstore.as_retriever(
    search_kwargs={"k": 5}
)

print("Retriever is ready.")
```

Retriever is ready.

```
query = "Which region had the highest revenue growth?"

retrieved_docs = retriever.invoke(query)

for i, doc in enumerate(retrieved_docs, start=1):
    print("=" * 80)
    print(
        f"Result {i} | "
        f"Page {doc.metadata.get('page')} | "
        f"Type: {doc.metadata.get('modality')}")
    print(doc.page_content[:700])
    print()
```

```
* **Lowest Value:** The Middle East is projected to have the lowest growth among the listed regions at 18%.
* **Main Trend:** The data indicates a clear hierarchy in growth
```

```
=====
Result 3 | Page 2.0 | Type: table
| Priority | 2026 target | Year-end status |
|:-----|:-----|:-----|
| European expansion | >25% YoY regional revenue growth | Achieved: 31% |
| Support speed | Average resolution <9 hours | Achieved in July; 8.1 hours in August |
| Facility utilization | 80%+ Penang utilization | Achieved: 82% |
| Product profitability | Flagship gross margin >44% | Achieved: AtlasEdge X1 at 46% |
=====
```

```
Result 4 | Page 4.0 | Type: text
NovaCore Systems Ltd. - Fictional company report created for a multimodal RAG demonstration
Page 4
```

```
51
4.5/5
Figure 2. Europe recorded the highest year-over-year regional growth. Asia Pacific, however, finished with the highest customer satisfaction score.
Europe benefited from demand in automotive inspection and warehouse robotics. North American growth was led by energy and cold-chain logistics. Asia Pacific maintained the strongest customer satisfaction result.
```

```
=====
Result 5 | Page 3.0 | Type: table
| Metric | FY2025 | FY2026 | Change |
|:-----|:-----|:-----|:-----|
| Revenue | $97.5M | $132.0M | +35.4% |
| Gross profit | $39.7M | $56.4M | +42.1% |
| Operating income | $13.8M | $24.3M | +76.1% |
| Operating margin | 14.2% | 18.4% | +4.2 pts |
| R&D investment | $12.1M | $17.9M | +47.9% |
```

```
rag_prompt = ChatPromptTemplate.from_template(
    """
    You are a helpful assistant answering questions about the
    NovaCore Systems FY2026 company report.

    Use ONLY the retrieved context below.

    If the answer is not available in the context, say:
    "I could not find that information in the report."

    Mention page numbers when possible.

    CONTEXT:
    {context}

    QUESTION:
    {question}

    ANSWER:
    """
)

text_rag_chain = (
    rag_prompt
    | text_llm
    | StrOutputParser()
)

print("Text RAG chain is ready.")
```

Text RAG chain is ready.

```
def format_context(retrieved_docs):
    parts = []

    for doc in retrieved_docs:
        page = doc.metadata.get("page")
        modality = doc.metadata.get("modality")

        parts.append(
            f"[Page {page} | {modality.upper()}]\n"
            f"{doc.page_content}"
        )

    return "\n\n".join(parts)
```

```
def answer_with_vision(question, context, image_paths):
    image_paths = image_paths[:3]

    content = [
        {
            "type": "text",
            "text": f"""
            You are answering questions about the NovaCore Systems FY2026 report.

            Use ONLY the retrieved context and the attached retrieved visuals.

            RETRIEVED CONTEXT:
            {context}

            QUESTION:
            {question}

            Instructions:
```

```

- Answer factually.
- Use the attached visuals when relevant.
- Mention page numbers.
- If the information is missing, say you could not find it.
"""
    }
]

for image_path in image_paths:
    content.append(
        {
            "type": "image_url",
            "image_url": {
                "url": image_to_data_uri(image_path)
            },
        }
    )

response = groq_client.chat.completions.create(
    model=VISION_MODEL,
    messages=[
        {
            "role": "user",
            "content": content,
        }
    ],
    temperature=0,
    max_completion_tokens=900,
)

return response.choices[0].message.content.strip()

```

```

def ask_rag(question, show_sources=True, show_images=False):

    # 1. Retrieve from Pinecone
    retrieved_docs = retriever.invoke(question)

    # 2. Create context
    context = format_context(retrieved_docs)

    # 3. Check whether Pinecone returned visuals
    image_paths = []

    for doc in retrieved_docs:
        if doc.metadata.get("modality") == "visual":
            image_path = doc.metadata.get("image_path")

            if image_path and Path(image_path).exists():
                image_paths.append(image_path)

    # 4. Generate the answer
    if image_paths:
        answer = answer_with_vision(
            question=question,
            context=context,
            image_paths=image_paths,
        )
    else:
        answer = text_rag_chain.invoke(
            {
                "context": context,
                "question": question,
            }
        )

    print("\nQUESTION:")
    print(question)

    print("\nANSWER:")
    print(answer)

    if show_sources:
        print("\nPINECONE RETRIEVED SOURCES:")

        for i, doc in enumerate(retrieved_docs, start=1):
            print(
                f"{i}. Page {doc.metadata.get('page')} "
                f"| {doc.metadata.get('modality')}"
            )

    if show_images and image_paths:
        print("\nRETRIEVED VISUALS:")

```

```
for image_path in image_paths:
    display(Image.open(image_path))

return {
    "answer": answer,
    "documents": retrieved_docs,
    "image_paths": image_paths,
}
```

```
ask_rag(
    "What does NovaCore Systems do and where is the company headquartered?"
)
```

QUESTION:

What does NovaCore Systems do and where is the company headquartered?

ANSWER:

NovaCore Systems Ltd. develops industrial AI gateways, computer-vision appliances, and connected sensor hubs for manufactu

PINECONE RETRIEVED SOURCES:

1. Page 6.0 | text
2. Page 1.0 | text
3. Page 8.0 | text
4. Page 2.0 | text
5. Page 5.0 | text

{'answer': 'NovaCore Systems Ltd. develops industrial AI gateways, computer-vision appliances, and connected sensor hubs for manufacturing, logistics, and energy companies [Page\u202f2.0] . The company is headquartered in Singapore [Page\u202f1.0] .',

'documents': [Document(id='8a1aa460-d944-46fc-83fd-b84fe5da15f3', metadata={'modality': 'text', 'page': 6.0, 'source': 'NovaCore_Multimodal_Company_Report_2026.pdf'}, page_content='NovaCore Systems Ltd. - Fictional company report created for a multimodal RAG demonstration\nPage 6\n5. Manufacturing & Supply Chain\nInformation-bearing diagram + facility image\nThe supply-chain diagram above contains operational details that are not repeated in full elsewhere. In particular, it\nidentifies the Quality Lab in Singapore as the critical control point before products move to regional distribution\nhubs.\nFigure 3. Penang Smart Assembly Facility. The embedded image carries facility utilization and solar contribution values that can be\nqueried by a multimodal system.'),

Document(id='3d5e9f59-29c5-4212-b96f-b59a98cddc8c', metadata={'modality': 'text', 'page': 1.0, 'source': 'NovaCore_Multimodal_Company_Report_2026.pdf'}, page_content='NovaCore Systems Ltd. - Fictional company report created for a multimodal RAG demonstration\nPage 1\nNOVACORE SYSTEMS LTD.\nFY2026 BUSINESS PERFORMANCE &\nOPERATIONS REPORT\nA realistic fictional company document designed for multimodal RAG testing\nHeadquarters\nSingapore\nEmployees\n1,480\n2026 Revenue\n\$132.0M\nOperating Margin\n18.4%\nPrimary Sector\nIndustrial AI & IoT\nMarkets Served\n42 countries\nDemo note: All company names, metrics, people, and events in this report are fictional. The document intentionally mixes text, tables,\ncharts, and embedded images so it can be used to demonstrate multimodal retrieval and question answering.'),

Document(id='374faa79-9e46-4470-a814-1c91434a6932', metadata={'modality': 'text', 'page': 8.0, 'source': 'NovaCore_Multimodal_Company_Report_2026.pdf'}, page_content='NovaCore Systems Ltd. - Fictional company report created for a multimodal RAG demonstration\nPage 8\n7. Sustainability & Facility Efficiency\nPie chart + sustainability metrics table\nFigure 5. Electricity mix at the Penang facility. Solar supplied 34% of electricity, purchased renewables 15%, and grid electricity 51%.\nMetric\n2025\n2026\n2027 target\nFacility electricity intensity\n118 kWh/unit\n101 kWh/unit\n92 kWh/unit\nProduction scrap rate\n3.8%\n2.9%\n2.5%\nPackaging recycled content\n52%\n68%\n80%\nScope 1+2 emissions\n8,950 tCO2e\n7,420 tCO2e\n6,600 tCO2e\nNovaCore plans to add rooftop solar capacity, upgrade burn-in test equipment, and increase recycled packaging\ncontent during FY2027. The company also intends to expand supplier carbon-data collection beyond its top 20\ncomponent vendors.'),

Document(id='88d70717-262d-4307-b3c2-a8d7653f108c', metadata={'modality': 'text', 'page': 2.0, 'source': 'NovaCore_Multimodal_Company_Report_2026.pdf'}, page_content='NovaCore Systems Ltd. - Fictional company report created for a multimodal RAG demonstration\nPage 2\n1. Executive Overview\nNarrative text section - useful for text retrieval and grounded summarization\nNovaCore Systems Ltd. develops industrial AI gateways, computer-vision appliances, and connected sensor hubs\nfor manufacturing, logistics, and energy companies. During FY2026, the company focused on three priorities:\nexpanding enterprise deployments in Europe, improving the gross margin of its flagship AtlasEdge product line,\nand shortening post-deployment support response times.\nFY2026 revenue reached USD 132.0 million, up from USD 97.5 million in FY2025. Europe became the\nfastest-growing region, supported by automotive and warehouse automation projects. The AtlasEdge X1 remained\nthe highest-margin product, while VisionHub Pro generated the largest number of new enterprise pilots.\nManagement approved a further USD 11.5 million investment in the Penang assembly facility for test automation\nand energy-efficiency upgrades.\nThe company ended the year with 1,480 employees. Engineering represented 38% of headcount, operations 27%,\nsales and customer success 22%, and corporate functions 13%. Customer satisfaction improved as average\nsupport resolution time declined steadily through August.\n2026 Strategic Priorities\nPriority\nYear-end status\nEuropean expansion\n>25% YoY regional revenue growth\nAchieved: 31%\nSupport speed\nAverage resolution <9 hours\nAchieved in July; 8.1 hours in August\nFacility utilization\n80%+ Penang utilization\nAchieved: 82%\nProduct profitability\nFlagship gross margin >44%\nAchieved: AtlasEdge X1 at 46%\nCross-modal clue: the exact electricity mix of the Penang facility appears later in a chart, while the\nsupply-chain control point appears inside an image diagram.'),

Document(id='45d136c4f5c0d3b4d30c1c5d44b0c5b1', metadata={'modality': 'text', 'page': 4.0, 'source': 'NovaCore_Multimodal_Company_Report_2026.pdf'}, page_content='2026 Year-over-Year Revenue Growth by Region\nBar chart showing revenue growth percentages for Europe, Asia, and North America. Europe shows the highest growth at 25%.\nTable with 2 columns: Region, Revenue Growth (%)\nEurope, 25%\nAsia, 18%\nNorth America, 12%'

```
ask_rag(
    "Which region had the highest year-over-year revenue growth?"
)
```

QUESTION:

Which region had the highest year-over-year revenue growth?

ANSWER:

Based on the bar chart titled "2026 Year-over-Year Revenue Growth by Region" and the accompanying table on **Page 4.0**, **E

PINECONE RETRIEVED SOURCES:

1. Page 4.0 | visual
2. Page 4.0 | table

```
3. Page 2.0 | table
4. Page 3.0 | visual
5. Page 3.0 | table
{'answer': 'Based on the bar chart titled "2026 Year-over-Year Revenue Growth by Region" and the accompanying table on
**Page 4.0**, **Europe** had the highest year-over-year revenue growth at **31%**.',
 'documents': [Document(id='75800ca2-ed2f-47af-a0f5-c19678ce6b7d', metadata={'image_path':
'novacore_extracted_images/page_4_image_1.png', 'modality': 'visual', 'page': 4.0, 'source':
'NovaCore_Multimodal_Company_Report_2026.pdf'}, page_content='Based on the visual provided, here is the description of the
business information:\n\n**Chart Type:** Bar Chart\n\n**Title:** 2026 Year-over-Year Revenue Growth by Region\n\n**Key Data
Points:**\n\nThe chart displays the projected revenue growth percentages for four specific geographic regions for the fiscal
year 2026:\n* **Europe:** 31%\n* **North America:** 27%\n* **Asia Pacific:** 24%\n* **Middle East:**
18%\n\n**Analysis:**\n* **Highest Value:** Europe is projected to have the highest year-over-year revenue growth at
31%.\n* **Lowest Value:** The Middle East is projected to have the lowest growth among the listed regions at 18%.\n\n
**Main Trend:** The data indicates a clear hierarchy in growth performance, with Europe leading, followed closely by North
America and Asia Pacific. All four regions are projected to experience positive double-digit growth.'],
 Document(id='7508dc5d-7c93-4433-a779-ad9d66e94f04', metadata={'modality': 'table', 'page': 4.0, 'source':
'NovaCore_Multimodal_Company_Report_2026.pdf', 'table_number': 1.0}, page_content='| Region | 2026 revenue | YoY
growth | Enterprise customers | CSAT | \n|:-----|:-----|:-----|:-----|:-----|:-----|:-----|:-----|:-----|:-----|
-----| \n| Europe | $41.2M | 31% | | | 118 | 4.7/5 | \n| North America | $38.6M
| 27% | | 104 | 4.6/5 | \n| Asia Pacific | $34.9M | 24% | | 96 | 4.8/5 | \n| Middle East | $17.3M | 18% | | 51 | 4.5/5 | '),
 Document(id='d0de4773-3b32-4832-89cc-7ba116e5fde', metadata={'modality': 'table', 'page': 2.0, 'source':
'NovaCore_Multimodal_Company_Report_2026.pdf', 'table_number': 1.0}, page_content='| Priority | 2026 target
| Year-end status | \n|:-----|:-----|:-----|:-----|:-----|:-----|:-----|:-----|:-----|:-----|
-----| \n| European expansion | >25% YoY regional revenue growth | Achieved: 31%
| \n| Support speed | Average resolution <9 hours | Achieved in July; 8.1 hours in August | \n| Facility
utilization | 80%+ Penang utilization | Achieved: 82% | \n| Product profitability |
Flagship gross margin >44% | Achieved: AtlasEdge X1 at 46% | '),
 Document(id='b42a5751-f7f8-4dfe-9b2e-d08656e856bf', metadata={'image_path':
'novacore_extracted_images/page_3_image_1.png', 'modality': 'visual', 'page': 3.0, 'source':
'NovaCore_Multimodal_Company_Report_2026.pdf'}, page_content='Based on the visual provided, here is the description of the
business information:\n\n**Chart Type:** Line Graph\n\n**Title:** Quarterly Revenue Trend\n\n**Key Data Points (Revenue in
USD millions):**\n* **Q1 2025:** ~22.0 (Lowest value)\n* **Q2 2025:** ~23.5\n* **Q3 2025:** ~25.2\n* **Q4 2025:**
~27.2\n* **Q1 2026:** ~28.5\n* **Q2 2026:** ~31.5\n* **Q3 2026:** ~34.1\n* **Q4 2026:** ~37.5 (Highest
value)\n\n**Main Trend:**\n\nThe chart displays a consistent and strong upward trend in revenue over the eight-quarter period
from Q1 2025 to Q4 2026. The company shows steady growth every quarter, with the total revenue increasing by approximately
15.5 million USD (from 22.0 to 37.5) over the two-year span. The growth appears to accelerate slightly in the second year
(2026) compared to the first.'],
 Document(id='2d1c86f7-d964-4b2a-8477-0efec56319da', metadata={'modality': 'table', 'page': 3.0, 'source':
'NovaCore_Multimodal_Company_Report_2026.pdf', 'table_number': 1.0}, page_content='| Metric | FY2025 | FY2026
| Change | \n|:-----|:-----|:-----|:-----|:-----| \n| Revenue | $97.5M | $132.0M | +35.4%
| \n| Gross profit | $39.7M | $56.4M | +42.1% | \n| Operating income | $13.8M | $24.3M | +76.1% | \n|
Operating margin | 14.2% | 18.4% | +4.2 pts | \n| R&D investment | $12.1M | $17.9M | +47.9% | ')],
 'image_paths': ['novacore_extracted_images/page_4_image_1.png',
'novacore_extracted_images/page_3_image_1.png']}]
```

```
ask_rag(
    "According to the revenue graph, which quarter had the highest revenue?",
    show_images=True,
)
```


QUESTION:

According to the revenue graph, which quarter had the highest revenue?

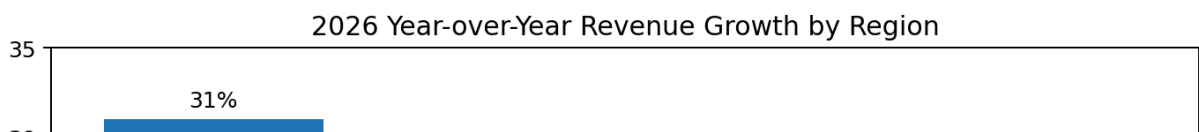
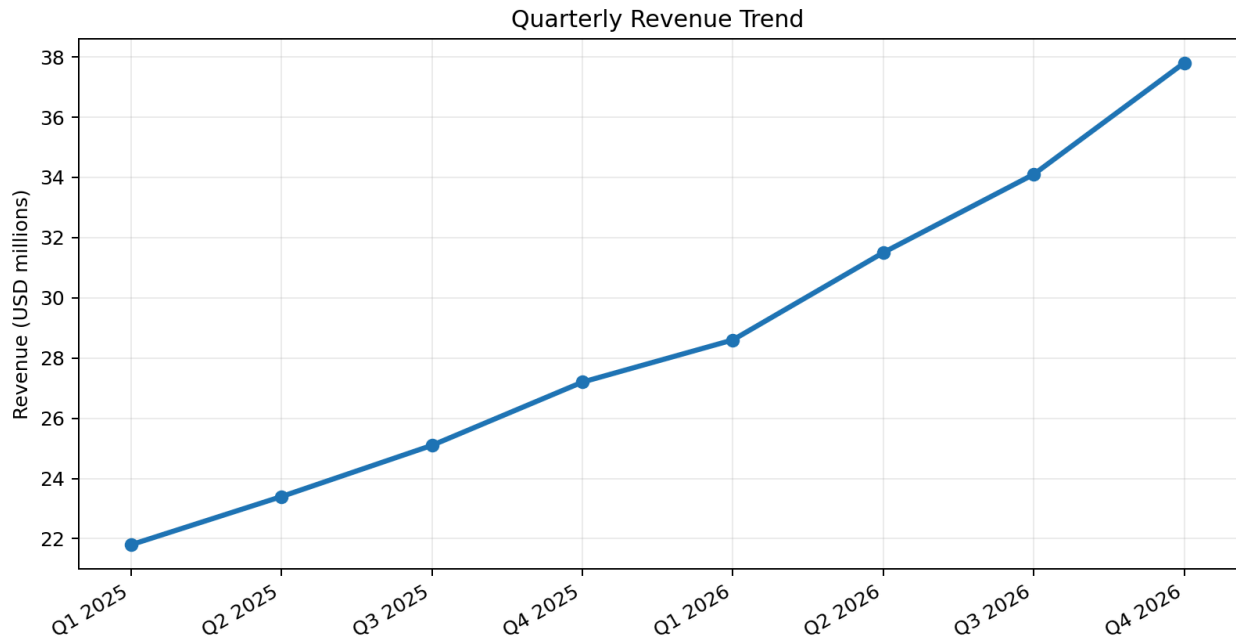
ANSWER:

Based on the "Quarterly Revenue Trend" line graph on Page 3.0, the quarter with the highest revenue is **Q4 2026**. The gr

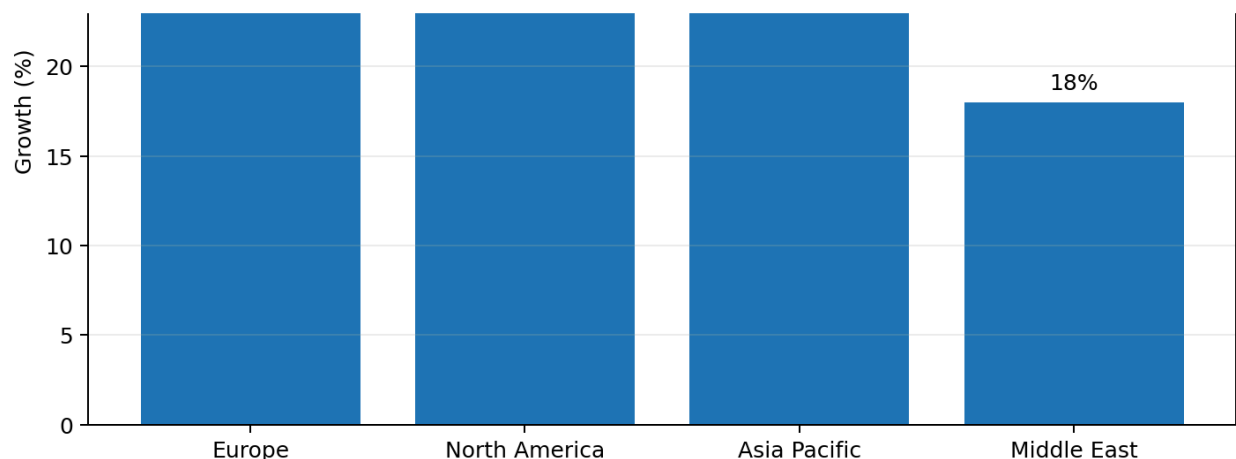
PINECONE RETRIEVED SOURCES:

1. Page 3.0 | visual
2. Page 4.0 | visual
3. Page 4.0 | table
4. Page 3.0 | table
5. Page 3.0 | text

RETRIEVED VISUALS:



```
ask_rag(  
    "According to the supply-chain diagram, what is the critical quality-control point?",  
    show_images=True,  
)
```



```
{  
  'answer': 'Based on the "Quarterly Revenue Trend" line graph on Page 3.0, the quarter with the highest revenue is Q4 2026. The graph shows a consistent upward trend, with Q4 2026 reaching the peak value of approximately $37.5 million.',  
  'documents': [  
    Document(id='b42a5751-f7f8-4dfe-9b2e-d08656e856bf', metadata={'image_path':  
      'novacore_extracted_images/page_3_image_1.png', 'modality': 'visual', 'page': 3.0, 'source':  
      'NovaCore_Multimodal_Company_Report_2026.pdf'}),  
    Document(id='75800ca2-ed2f-47af-a0f5-c19678ce6b7d', metadata={'image_path':  
      'novacore_extracted_images/page_4_image_1.png', 'modality': 'visual', 'page': 4.0, 'source':  
      'NovaCore_Multimodal_Company_Report_2026.pdf'})  
  ],  
  'page_content': 'Based on the visual provided, here is the description of  
the business information:\n\n**Chart Type:** Line Graph\n\n**Title:** Quarterly Revenue Trend\n\n**Key Data Points (Revenue in USD millions):**\n\n    **Q1 2025:** ~22.0 (Lowest value)\n    **Q2 2025:** ~23.5\n    **Q3 2025:** ~25.2\n    **Q4 2025:** ~27.2\n    **Q1 2026:** ~28.5\n    **Q2 2026:** ~31.5\n    **Q3 2026:** ~34.1\n    **Q4 2026:** ~37.5 (Highest value)\n\nThe chart displays a consistent and strong upward trend in revenue over the eight-quarter period from Q1 2025 to Q4 2026. The company shows steady growth every quarter, with the total revenue increasing by approximately 15.5 million USD (from 22.0 to 37.5) over the two-year span. The growth appears to accelerate slightly in the second year (2026) compared to the first.'  
Based on the visual provided, here is the description of
```

